

Assignment 4
SWE 432 Fall 2024
REVISED

Total Points: 150

1. Design (30 points)

- Initial Component Diagram (10 points):

- Developed an initial component diagram prior to coding showing an appropriate architecture for an Node.JS/EJS implementation in a relative MVC style. A short explanation must be provided that documents the components.

- Initial Sequence Diagram (10 points):

- Developed an initial sequence diagram prior to coding showing the module flow. A short explanation must be provided that explains the flow.

- Updated Diagrams (10 points):

- Updated the diagrams to match the delivered system

2. EJS Implementation (50 points)

- EJS Syntax (20 points):

- Example: The student demonstrates a good understanding of EJS syntax

- Data Passing (10 points):

- Passing data from their server to their EJS templates successfully.

- EJS Views (10 points):

- Created dynamic EJS views for their radio station application, utilized EJS layouts to create all pages.

Directory Structure (10 points)

- They have organized their project directory properly, separating routes, views, and static files.

3. Functionality (20 points)

- All features of the application are working as expected. Links lead to the correct pages, buttons perform their intended actions, and data is displayed correctly

4. Database Implementation (30 points)

- Database Setup (10 points):

- Successful setup of MongoDB, including installation, configuration, and connection to the web

application.

- Data Model Schema Design (10 points):

- Development of a MongoDB schema that accurately represents the data model for the application's needs, such as songs, DJs, playlists, listener preferences, etc.

- Frontend and Backend Data Sync (10 points):

- Complete migration of existing data to the new database structure, ensuring that all previously hard-coded data is now stored and retrieved from the database
- Verification that frontend displays are updated to reflect the backend data accurately after CRUD (Create, Read, Update, Delete) operations, ensuring data consistency across the application.

5. Session Management and Continuity (20 points)

- Data Persistence and Restoration (10 points):

- Ensuring that session data is persistent across page reloads and that the correct state of the user's interaction is maintained, except for in-process actions.

E.g.- Describe a scenario where a listener is creating a list of favorite songs and, after a page reload, the list remains intact, demonstrating the session's persistence.

- Logout and Session Clearing (10 points):

- Effective implementation of a logout mechanism that clears the current session while preserving the user's data integrity in the database.